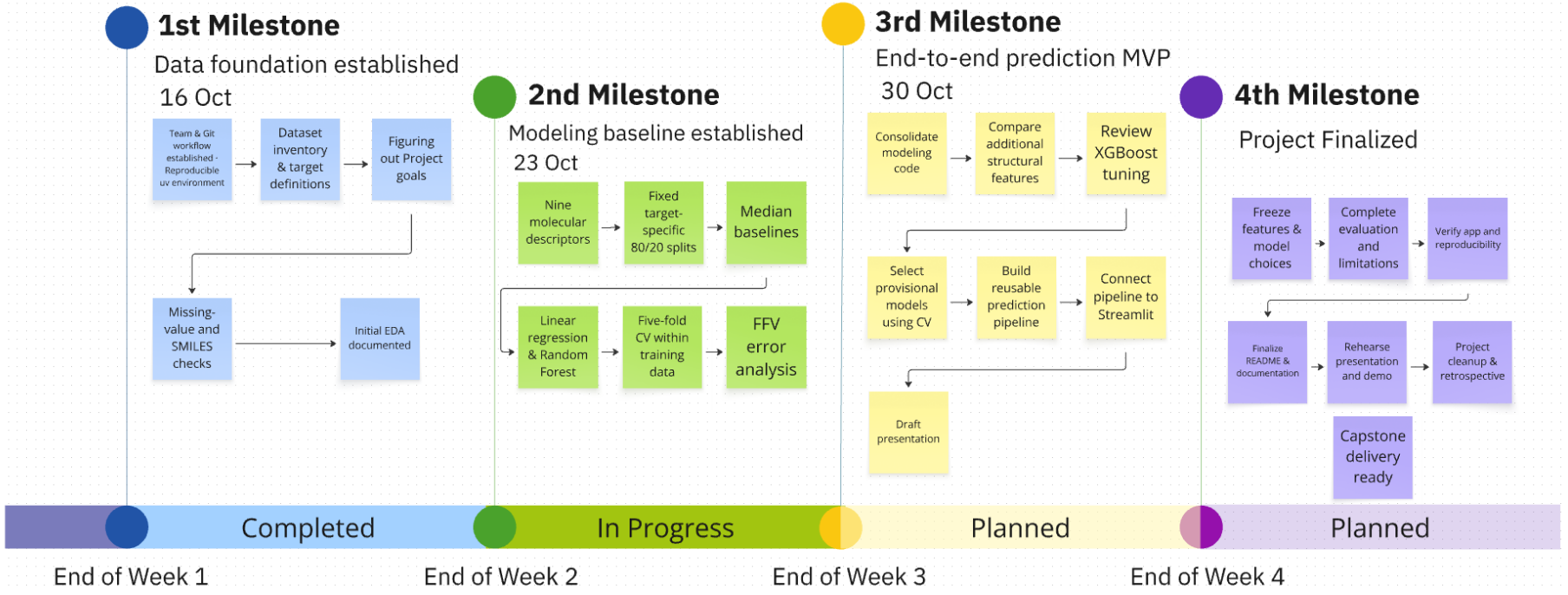
Team Polymind 😎

Data-Driven Materials Informatics: Machine Learning for sustainable Polymer Property Prediction

Our Timeline



Five targets with very different amounts of data

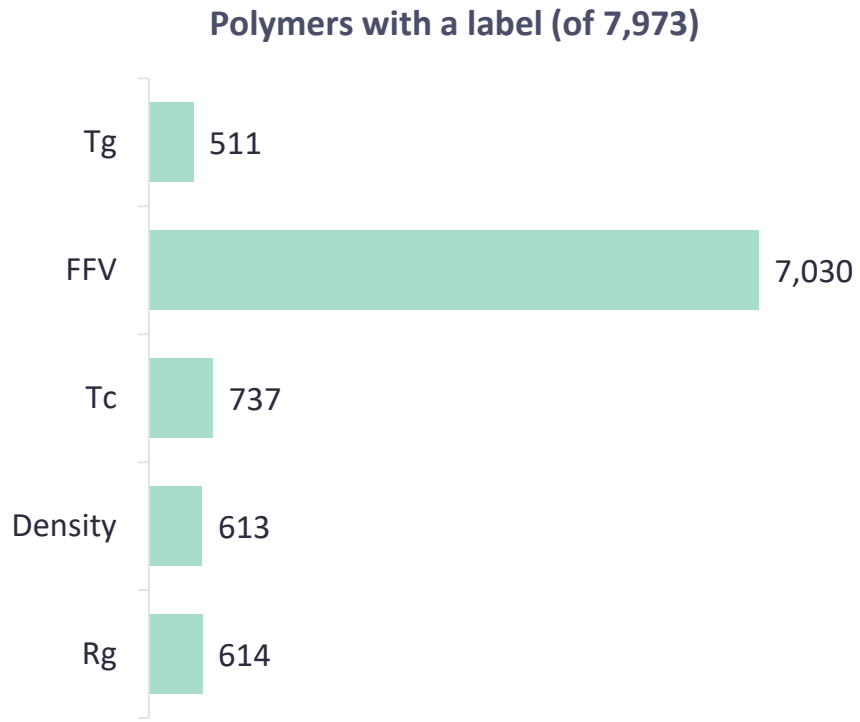
Tg
Glass transition temperature · °C
Where the amorphous polymer changes from glassy to rubbery

FFV
Fractional free volume · dimensionless
Share of volume not occupied by chains; governs gas permeability

Tc
Thermal conductivity · W/(m·K)
How well heat is transported through the material

ρ
Density · g/cm³
Mass per volume; reflects atom masses and chain packing

Rg
Radius of gyration · Å
Average spatial size of a polymer chain



No polymer has all five labels
→ one separate model per target. Tg, Density and Rg have only ~500–600 labels each.

Labels from molecular-dynamics simulations · Kaggle NeurIPS Open Polymer Prediction 2025 (train.csv)

Our goal: fill the gaps, but only where we can trust it

Today: 7,973 polymers, none has all five labels

Polymer	Tg	FFV	Tc	Dens.	Rg
#1	?	✓	?	?	?
#2	?	✓	?	?	?
#3	✓	?	?	?	?
#4	?	✓	✓	✓	✓
#5	?	✓	?	?	?
#6	?	?	✓	✓	✓
#7	?	✓	?	?	?
#8	✓	✓	?	?	?
Labels	**511**	**7,030**	**737**	**613**	**614**

Goal: predict the missing values

Tg	FFV	Tc	Dens.	Rg
≈	✓	≈	≈	≈
≈	✓	≈	≈	≈
✓	≈	≈	≈	≈
≈	✓	✓	✓	✓
≈	✓	≈	≈	≈
≈	≈	✓	✓	✓
≈	✓	≈	≈	≈
✓	✓	≈	≈	≈

✓ measured label

? missing (94% for Tg)

≈ model prediction

1 **Predict all five**

- from the SMILES of a repeat unit

2 **≥ 40% smaller error**

- than always guessing the median

3 **Fill gaps only if reliable**

- Tg is not there yet

Labels per target in train.csv (7,973 polymers). Error reduction compared with always predicting the median. Illustration: 8 example rows.

How we evaluate: a frozen test set, choices by cross-validation

1

Frozen validation set

- 20% of each target's labels, split once (seed 42)
- Never used for any choice
- Scored once at the end per model version

2

5-fold cross-validation

- On the other 80% (training polymers)
- Every feature, model and setting chosen here
- Same folds for every comparison

3

Metric and goal

- MAE in the unit of each property
- Error reduction = 1 − MAE / MAE of always predicting the median
- Goal: ≥ 40% error reduction

Why: scores on data used for decisions are optimistic. Only the frozen set shows how the model does on polymers it has never seen.

Training sizes after the split: Tg 408 · FFV 5,624 · Tc 589 · Density 490 · Rg 491 polymers.

Our features: from 9 to 43 descriptors

Original 9 (start)

Size, polarity, rings

MolWt mass of the repeat unit

NumAtoms, HeavyAtomCount size

NumRings, NumAromaticRings rings

NumHDonors, NumHAcceptors H-bonding

MolLogP hydrophobicity

TPSA polar surface area

+12 chemistry (set B)

Flexibility, composition, backbone

NumRotatableBonds chain flexibility

FractionCSP3, AromaticAtomFraction aliphatic vs. aromatic

Halogen, S, Si counts heavy atoms, Si–O backbones

MolWt, MolLogP, TPSA per heavy atom size-independent mass and polarity

BackboneLength atoms in the main chain

BackboneRingFraction rings in the backbone vs. pendant

SideChainFraction share of atoms outside the backbone

+22 functional groups (set B+)

Where a group sits matters

Ester, carbonate, amide, imide, urethane, ether, siloxane, sulfone each counted in backbone and in side chain (16)

Hydroxyl, nitrile side chain only (2)

LargestSideGroup long side chains plasticise

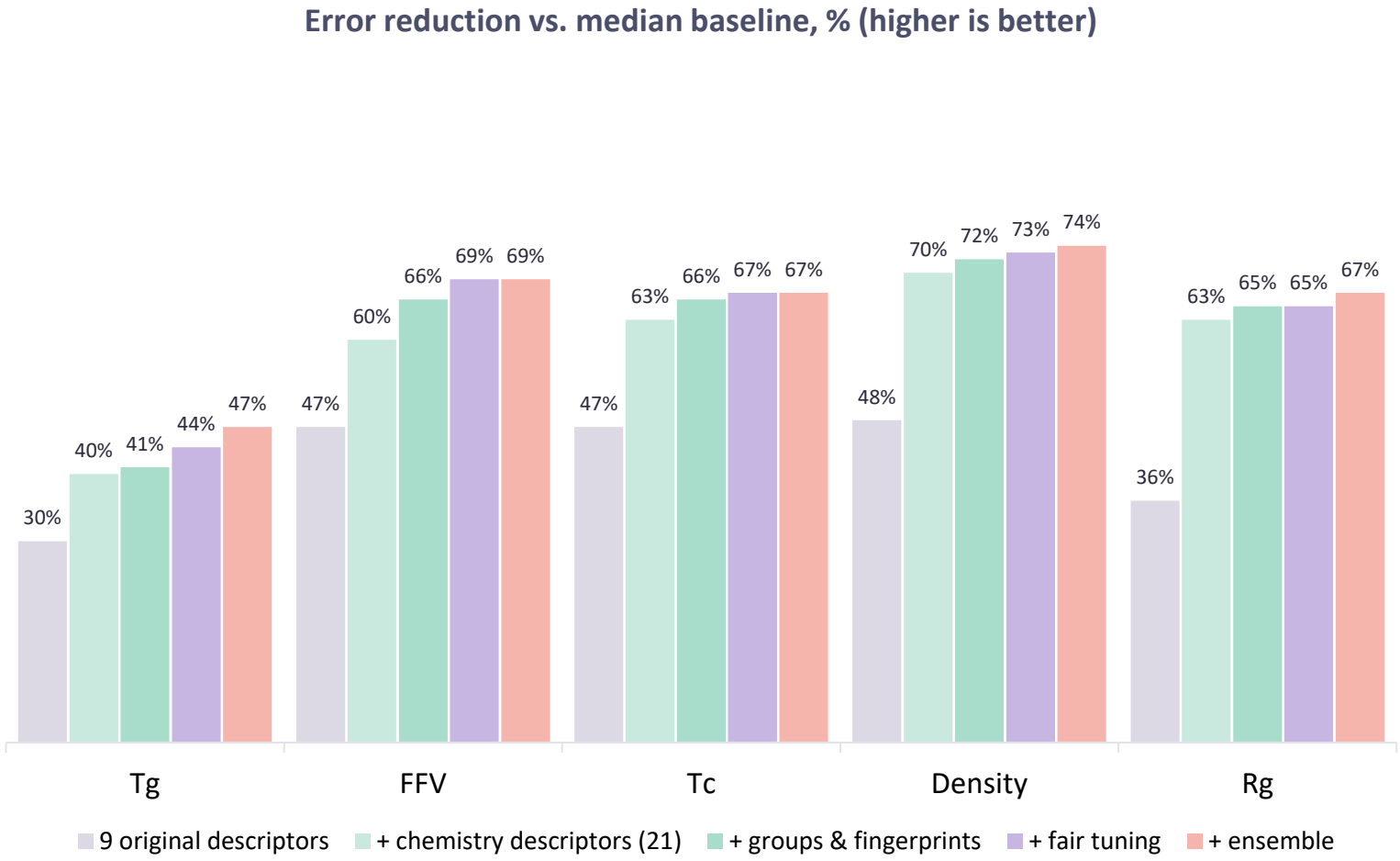
BackboneUnsaturation C=C in the main chain

H-donors, H-acceptors per heavy atom H-bonding density

Example: PET has its ester in the backbone, PMMA in the side chain

*Feature set B+ = all 43 descriptors. Optionally + 1,024 Morgan fingerprint bits (local atom neighbourhoods up to two bonds). Backbone = shortest path between the two * attachment points.*

Where the improvement came from



Takeaways

- The chemistry descriptors gave by far the biggest jump (Density 48% → 70%, Rg 36% → 63%).
- Every later step added only +0 to +7 points.
- All five targets now pass our 40% goal in cross-validation (Tg 47%).
- Supplementary Kaggle data: tested, no gain (FFV +1 point, Tg and Tc unchanged).
- **Features matter more than model choice.**

Error reduction = 1 − MAE / MAE of the median baseline, best model per step, 5-fold cross-validation. Final step: ensemble for Tg, Tc, Density, Rg; tuned XGBoost for FFV.

Fair tuning: XGBoost wins, but the gain is small

Target	Random forest, default	XGBoost, default	XGBoost, tuned	Gain from tuning	Tuned XGBoost MAE
Tg	41%	41%	**44%**	+3	49 °C
FFV	65%	66%	**69%**	+3	0.0063
Tc	66%	66%	**67%**	+1	0.025 W/(m·K)
Density	68%	72%	**73%**	+1	0.029 g/cm³
Rg	64%	65%	**65%**	+0	1.37 Å

✓ With the same tuning budget, XGBoost wins for every target.

✓ The default settings were already close to optimal: tuning adds only 0 to 3 points.

Error reduction vs. median baseline (%), 5-fold cross-validation, best of B+ and B+ + D. Tuning: 30 random settings per model, same folds; settings were selected by R^2 (not yet re-run with MAE).

One model for all five targets, without filling the gaps

How it works: the masked loss

Polymer	Tg	FFV	Tc	Dens.	Rg
A	–	✓	–	–	–
B	–	–	✓	✓	✓
C	✓	–	–	–	–

- One network, five outputs.
- Each polymer trains only the outputs it has real values for (✓).
- Missing values (–) are skipped, not filled: the model never learns from guesses.
- Learns from the real labels of all five targets at once.

Target	Multi-task NN (1 model)	Single-task NN (5 models)	Tuned XGBoost (5 models)
Tg	**43%**	40%	44%
FFV	62%	66%	69%
Tc	**64%**	62%	67%
Density	70%	69%	73%
Rg	**63%**	57%	65%

✓ Sharing helps the small targets: Tg +3, Tc +2, Rg +6 points vs. separate networks.

~ FFV loses 4 points; XGBoost is still best overall.

✗ Fingerprints overfit the network (too many inputs for ~500 labels).

Table: error reduction vs. median baseline (%), 5-fold cross-validation. Neural networks: 2 hidden layers (512, 256), not tuned; best feature set per model. Single-task: same architecture, one per target.

Final check on unseen data: errors 65–75% smaller, Tg 33%

Target	Final model	MAE: median baseline	MAE: starting model	MAE: final model	Error reduction	≥ 40% goal
Tg	Ensemble	78.7 °C	62.6 °C	**52.4 °C**	33%	✗
FFV	Tuned XGBoost	0.0214	0.0109	**0.0062**	71%	✓
Tc	Ensemble	0.080 W/(m·K)	0.039 W/(m·K)	**0.023 W/(m·K)**	72%	✓
Density	Tuned XGBoost	0.104 g/cm³	0.066 g/cm³	**0.027 g/cm³**	75%	✓
Rg	Ensemble	3.86 Å	2.87 Å	**1.35 Å**	65%	✓

- In plain words: density within ±0.03 g/cm³ (about 3%), Rg within ±1.4 Å, Tg within ±52 °C on average.
- Compared with the starting model on the same polymers, the error drops by 16% (Tg) to 60% (Density).
- **Not yet included:** external Tg data and Chemprop (slides 12–14). They get one final check after integration.

Validation subsets = 20% of each target's labels, never used for any decision. Final models use 73 descriptors + fingerprints. Ensemble = 50/50 multi-task NN + tuned XGBoost.

The demo runs five saved models through one interface

Target	Model	Structure representation
Tg	Random forest	Morgan 1,024 + 21 descriptors
FFV	Random forest	Morgan 1,024
Tc	XGBoost	21 descriptors
Density	XGBoost	21 descriptors
Rg	Random forest	Morgan 1,024

Our current working version

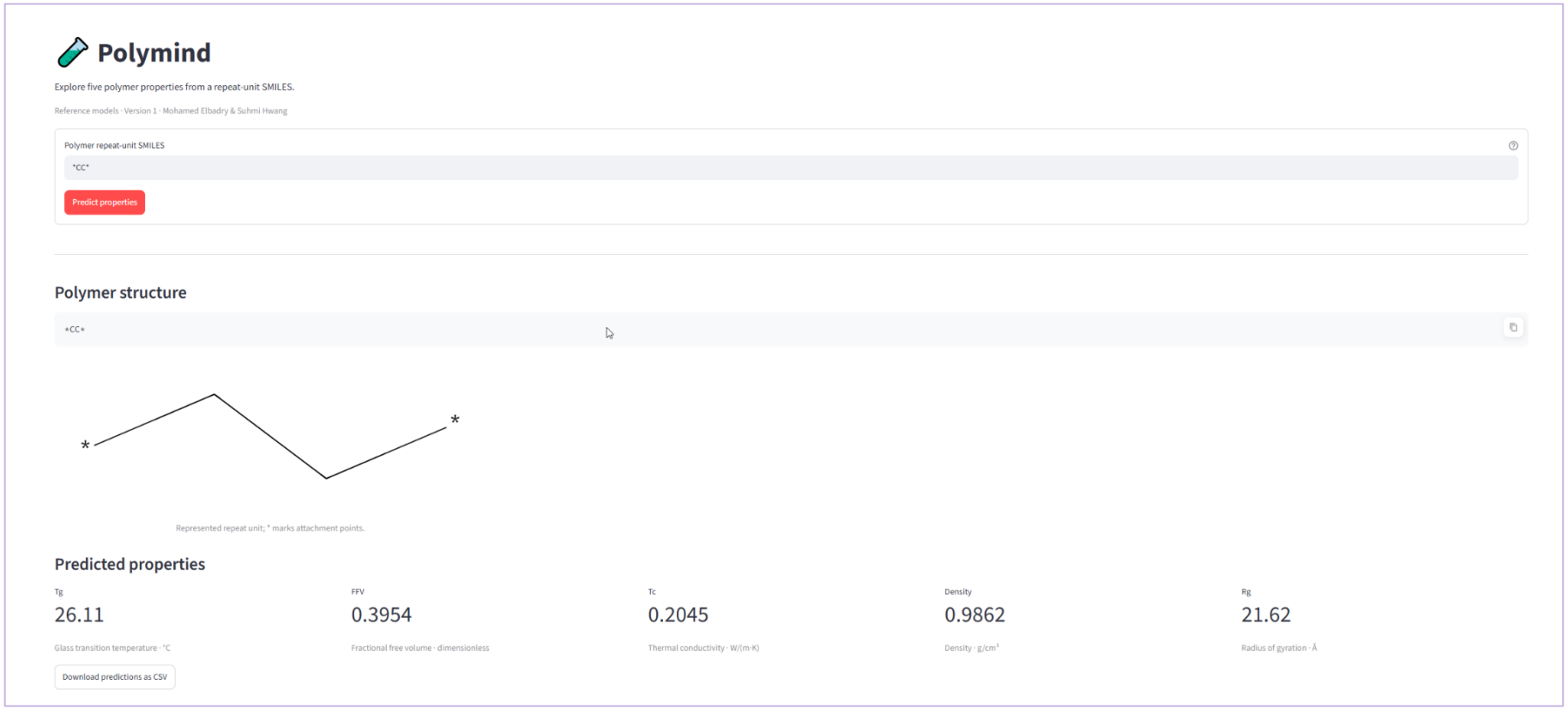
- One function returns all five properties
- Fixed, saved and documented, with its own holdout check (backup slide)
- Simple, robust models per target

Next

- Swap in the improved models once they are verified on the same split

Separate model per target. Morgan fingerprints: radius 2, chirality enabled. Full settings: backup slides.

From model to tool: SMILES in, five properties out



Use case: rank and pre-screen candidate polymers before simulations or lab tests, not replace them.

*Polymind Streamlit demo, example *CC* (polyethylene). Model estimates, not experimental measurements.*

Why Tg is the hardest target: noisy labels and extremes

±30 °C
same polymer, different literature sources: label noise

0.69
slope of predicted vs. true Tg: predictions pulled to the mean

61 / 60 °C
error for the lowest / highest Tg fifth (middle: 34 °C)

58 vs 38 °C
error for polymers unlike vs. like the training data

Tested, no gain

- **Clustering** (coach's suggestion): weak clusters (silhouette 0.13); per-cluster models not better
- **Removing suspicious labels** (9 fully aromatic polymers with Tg < 0 °C): 48.2 → 47.9 °C
- **Focusing on extremes** (weights, MAE objective, less regularisation): no gain
- **Pretrained Uni-Mol embeddings:** worse (Tg 52.1 vs 48.2 °C)

→ **The limit is information, not the model:** we need more Tg data.

Tg error analysis, 5-fold cross-validation on the 408 Tg training polymers (notebook 08).

More data helps Tg: external dataset cuts the error 48 → 44 °C

7,367
LAMALAB curated experimental Tg

→

7,231
reliable entries (quality flags)

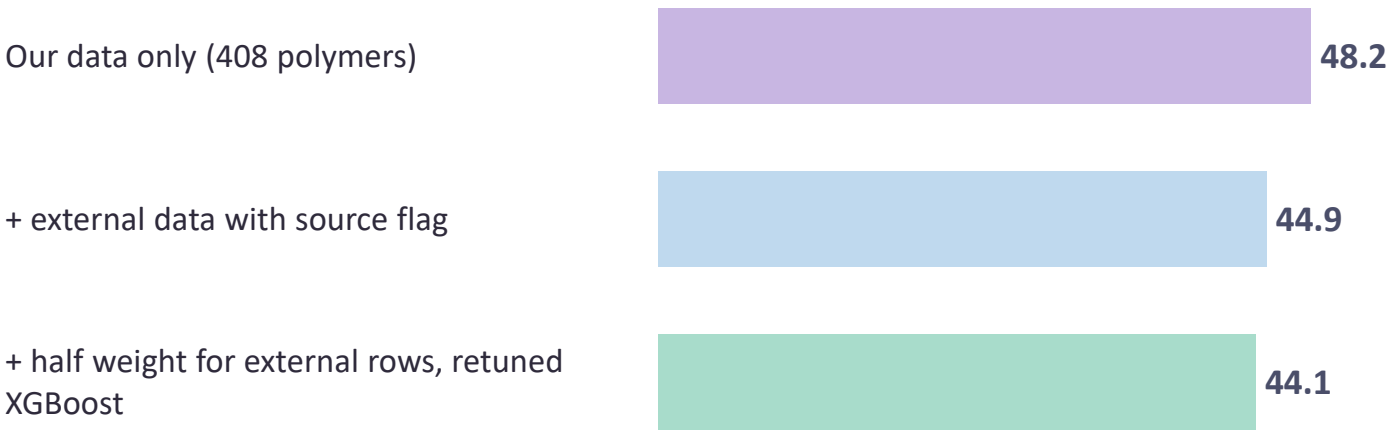
→

7,225
unique structures

→

7,194
after removing all overlaps with our data

Tg cross-validated MAE (°C, lower is better)



Checks

- Tg in K → °C; experimental values
- Overlap removed by normalised structure (+ similarity ≥ 0.95 to validation polymers)
- Source flag absorbs a ~20 °C offset between datasets
- **Validation set untouched; experimental, not yet integrated**

Source: Kunchapu & Jablonka, LAMALAB curated Tg dataset, Zenodo 15210035, CC-BY 4.0. Error reduction 45.6% → 50.2% (5-fold CV).

Latest experiments: what helped and what did not

Experiment	Target	Result (5-fold CV)	Verdict
External Tg data (LAMALAB)	Tg	MAE 48.2 → 44.1 °C; error reduction 46% → 50%	✓ clear gain
Graph neural network (Chemprop) + XGBoost	FFV	Error reduction 72% → 77% (MAE 0.0058 → 0.0048)	✓ clear gain
Graph neural network (Chemprop) + XGBoost	Tc	Error reduction 67% → 69%	~ within noise
Chain + 3D shape descriptors (73 in total)	all	+0 to +3 points	~ small gain
XGBoost re-tuned for MAE	all	Only FFV changed (+1.6 points)	~ small gain
Pretrained Uni-Mol embeddings	all	Worse for every target	✗ no gain
Interaction, size and packing descriptors	Tg, Rg, Density	No change (information already covered)	✗ no gain
Kaggle supplementary data	Tg, FFV, Tc	Mostly duplicates of our data; no gain	✗ no gain

All results on training polymers only; the validation set was not rescored. Experimental results, not yet in the current models.

Where we stand and what comes next

Target	Best model so far (CV)	Error reduction
Tg	XGBoost + external data	50% (44.1 °C)
FFV	Chemprop + XGBoost	77%
Tc	Chemprop + XGBoost	69%
Density	Tuned XGBoost	79%
Rg	Multi-task NN + XGBoost	69%

All five targets pass the 40% goal. Features and data matter more than the model; Tg is limited by label noise.

Next steps

- Review and verify the external-data Tg model together
- Integrate the new Tg (and FFV) model with its feature calculation into the prediction interface
- One final check of the updated models on the frozen validation set
- Finalise demo, documentation and final presentation

Best = lowest 5-fold CV MAE on the training polymers. Holdout results of the final models: slide 9.

Backup: settings of the five saved models

Target	Algorithm	Fixed configuration
Tg	Random forest	200 trees, max_features 0.6, min_samples_leaf 2
FFV	Random forest	200 trees, default settings
Tc	XGBoost	500 trees, learning_rate 0.05, max_depth 4
Density	XGBoost	500 trees, learning_rate 0.05, max_depth 4
Rg	Random forest	200 trees, default settings

All models: seed 42. XGBoost subsample 0.8 and colsample_bytree 0.8. Full settings: models/polymind_v1/manifest.json.

Backup: holdout metrics of the five saved models

Target	n	MAE	RMSE	R^2	MAE: median baseline
Tg	87	45.9 °C	58.8 °C	0.67	79.6 °C
FFV	1,419	0.0079	0.0121	0.80	0.0199
Tc	145	0.034 W/(m·K)	0.049 W/(m·K)	0.69	0.074 W/(m·K)
Density	123	0.034 g/cm³	0.074 g/cm³	0.72	0.092 g/cm³
Rg	124	1.84 Å	2.81 Å	0.67	4.12 Å

Measured on the shared reference split, a different holdout set than slide 9, so the numbers are not directly comparable.

Source: results/08_selected_model_validation/. Random holdout results, not leaderboard scores.